Development of a Common Outcome Data Set for Fall Injury Prevention Trials: The Prevention of Falls Network Europe Consensus

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The prevention of injury associated with falls in older people is a public health target in many countries around the world. Although there is good evidence that interventions such as multifactorial fall prevention and individually prescribed exercise are effective in reducing falls, the effect on serious injury rates is unclear.[1,2] Historically, trials have not been adequately powered to detect injury endpoints, and variations in case definition across trials have hindered meta-analysis.[1] It is possible that fall-prevention strategies have limited effect on falls that result in injuries or are ineffective in populations who are at a higher risk of injury. Further research is required to determine whether fall-prevention interventions can reduce serious injuries.

Prevention of Falls Network Europe (ProFaNE) is a collaborative project to reduce the burden of fall injury in older people through excellence in research and promotion of best practice (www.profane.eu.org). The European Commission funds the network, which links clinicians, members of the public, and researchers worldwide. The aims are to identify major gaps in knowledge in fall injury prevention and to facilitate the collaboration necessary for large-scale clinical research activity, including clinical trials, comparative research, and prospective meta-analysis. Work is being undertaken in a 4-year program. As a first step, the development of a common set of outcome definitions and measures for future trials or meta-analysis was considered. **J Am Geriatr Soc 53:1618–1622, 2005.**

Key words: accidental falls, aged, fractures, consensus development, outcome measures

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METHODS

Recommendations were generated by international expert consensus in three phases (summarized in Figure 1).

1. An international meeting of 32 experts in which agreement on the key domains for outcome assessment was secured.
2. Within each domain, systematic literature reviews were used to identify the quality and scope of measures used in clinical trials to date.
3. A further 2-day international meeting in which expert consensus recommendations for outcome measures in each domain were developed.

Full methodological details and results of searches are available.[3] Literature searches to identify measures used in clinical trials were an exact replication of the Cochrane Review of Interventions to Prevent Falls in the Elderly.[1] Secondary searches identified additional reports of the measures. All measures were assessed for quality in five areas—case definition, reliability, validity, responsiveness, and acceptability—and the results were summarized in a series of briefing papers. A group of 42 representatives of the academic, policy, practice, and user communities attended a consensus conference in November 2003. The group was drawn from a range of disciplines and continents, including Australasia, Europe, and North America. The consensus building used a modified nominal group technique[4] that included integrating evidence from the briefing papers alongside expert opinion and member checking to approve the final consensus statements. Discussions, which included the rationale for decisions, were taped, transcribed, and summarized.

Recommendations

Recommendation 1: Domains and Considerations

1. Domains should include falls, fall injury, physical activity, psychological consequences, and generic health-related quality of life (HRQoL).
2. The selection of measures should focus on community-dwelling populations.

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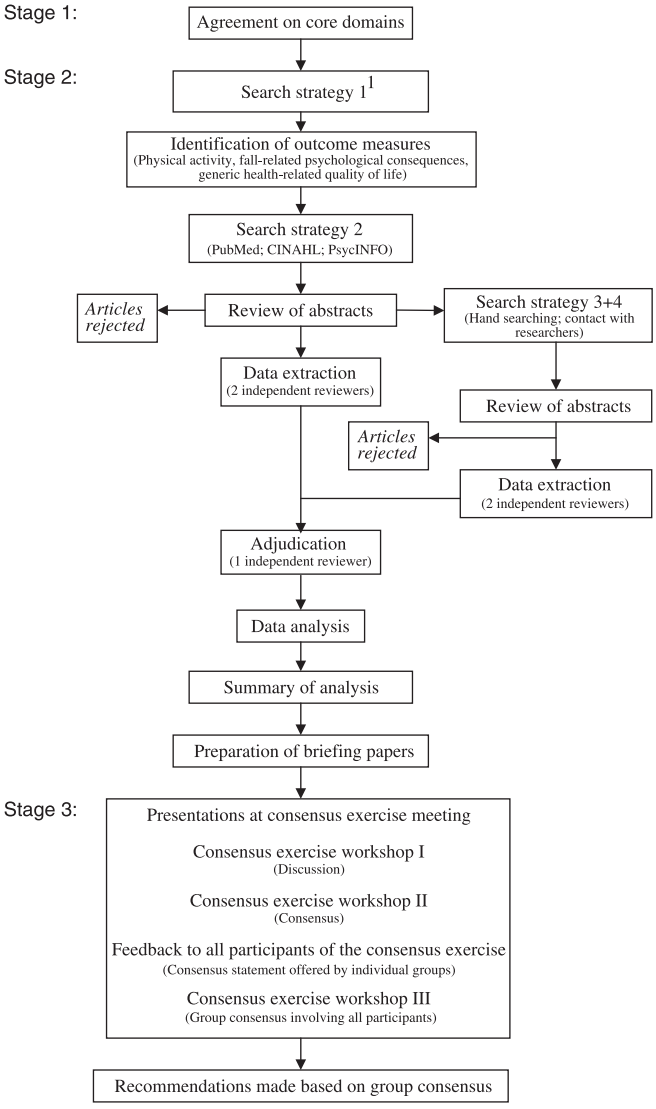


Figure 1. Flow chart of the consensus process.

3. The common data set should consider cost and ease of application in a wide range of countries.
4. The recommendations should include details on methods of measurement.
5. The process should be founded on a review of measures currently reported in clinical trials of fall and fall injury prevention interventions.

Rationale: The inclusion of falls and injuries is not surprising. The psychological domain was included because it is an independent risk factor for poor outcome after fall-prevention intervention[5] and is thus a legitimate and important target for interventions. Likewise, physical activity is an important outcome, enabling autonomy and independence. The group felt that the desirable profile of an intervention is one that improves activity and confidence while reducing falling. HRQoL is likely to capture unanticipated effects of an intervention in terms of general, emotional, and social health.[6]

The decision to focus on community-dwelling populations was a practical consideration because time and funding was limited. Future revisions of the consensus statement will consider whether the recommendations are generalizable to institutional settings.

The consensus was that the size of the common outcome data set should be carefully balanced to minimize the burden on respondents and maximize follow-up.[7] Although smaller trials and those conducted in more affluent countries might afford a greater number of and more-intensive measurement protocols, the common data set proposed should be achievable in most contexts. The common data set is intended to promote consistency in collection and reporting of essential elements. It is not the intention to suggest that investigators should be limited in using additional outcomes if indicated.

Domains rejected from the outcome data set included cognitive status and depression. Although these provide descriptive information and might be important baseline covariates, neither was considered a primary target of fall-prevention interventions. It was considered that judicious selection of HRQoL outcomes could detect general mental health effects.

Recommendation 2: Falls

1. A fall should be defined as "an unexpected event in which the participants come to rest on the ground, floor, or lower level."
2. Ascertainment must consider the lay perspective of falls. Participants should be asked, "In the past month, have you had any fall including a slip or trip in which you lost your balance and landed on the floor or ground or lower level?"
3. Falls should be recorded using prospective daily recording and a notification system with a minimum of monthly reporting. Telephone or face-to-face interview should be used to rectify missing data and to ascertain further details of falls and injuries.
4. Fall data should be summarized as number of falls, number of fallers/nonfallers/frequent fallers, fall rate per person year, and time to first fall (as a safety measure).
5. Primary analysis of fall data should not be adjusted for physical activity, and reporting should include the absolute risk difference between groups.

Rationale: The systematic review[3] revealed large variation in the definition of fall events and reinforced the need for standardization. The group felt that this variation was likely to be a major factor explaining differences in fall rates in studies conducted in similar populations.[8] A simple definition was deemed to be more appropriate for a common data set than definitions that specified medical events or situations (such as the Kellogg definition[9]). The recommendation does not preclude investigators from reporting, in addition, falls classified into subgroups based on etiology or context.

The literature on difficulties of collecting fall data was considered. Recall can be problematic.[10,11] Although older people are able to recall falls in a general way over a 1-year period, recall for events within the previous 3 to 6 months is less accurate. Prospective registration systems, in which a daily record of falls is made, are used widely. Monthly telephone follow-up has proven to be effective,[8,12–15] the purpose being to confirm that the event being reported fits the definition of a fall and to document resultant injuries.

Some researchers advocate weekly telephone follow-up,[10,11] others monthly,[13] but the difference between these is not known. Motion devices were not considered to be feasible.

The consensus panel recognized variations in the interpretation and meaning of falls across cultures. For example, there is no literal translation for the term "fall" in German. Also, older people and health professionals differ in their perspectives of falls.[16] The lay definition proposed covers a range of possible interpretations of fall events. Data summary and reporting problems apply equally to fall and injury events. Data summaries should be simple. Methods of analysis should account for multiple events within individuals and require a thorough understanding of the distribution of events.

Recommendation 3: Injuries

1. The recommended common data set measure is the number of radiologically confirmed peripheral fracture events per person year. This should include the limbs and limb girdles.
2. Injuries should be classified according to the International Classification of Diseases, 10th Revision, classification system.
3. Data should be collected prospectively, alongside and using the same methods as for fall reporting.
4. Injury data should be summarized as peripheral fracture rate per person-year of follow-up, number of peripheral fractures, number of people sustaining peripheral fractures, and number of people sustaining multiple events.
5. Primary analysis should not be adjusted for physical activity, and reporting should include the absolute risk difference between groups.

Rationale: Despite pharmaceutical and medical advances, peripheral fractures remain a major threat to independence for older people. Peripheral fracture rate, verified using radiological evidence, was considered to be the only robust and feasible measure of injury that could be recommended. Peripheral fractures account for the majority of cost, morbidity, and mortality generated by fall-related injury.[17] Soft-tissue and organ injuries have been defined and classified in a range of ways, none of which were considered satisfactory. Injury severity scored according to resource use (such as number of injuries requiring attendance at the emergency department) was rejected because variability in treatment protocols and service configurations within and between countries render this liable to inaccuracy. Drawing a distinction between major and minor soft-tissue injuries has historically been problematic, and there is no widely accepted approach. Accurate soft-tissue injury classification would require rapid assessment of the injury—in terms of self-report or independent examination. Self-report was considered to be susceptible to inaccuracy, and in many situations it would not be feasible for an examination to be scheduled within the short recovery period of soft-tissue injuries. Vertebral fractures are not generally a consequence of falling and were hence excluded from the recommendation.

Recommendation 4: Psychological Consequences of Falling

1. Psychological consequences of falls should be conceptualized in terms of fall-related self-efficacy, defined as "the degree of confidence a person has in performing common activities of daily living without falling" and measured using the modified Falls Efficacy Scale (mFES).[18]
2. The measure should be scored per published guidance.[18]

Rationale: The group agreed that an increasing body of evidence suggests that many older people experience psychological difficulties related to falls.[19] The systematic review highlighted a lack of consistency in defining the psychological consequences of falling.[19] Constructs described commonly in the literature included self-efficacy, fear of falling, and activity limitation. The consensus was that self-efficacy was the most appropriate construct, because it is developed from a strong theoretical base in social cognitive theory.[20]

The mFES was selected as the measure of self-efficacy for the common outcome data set because it has good test–retest reliability[18] and adequate levels of internal consistency[18] and having been used in several trials, has proven feasible, acceptable, and practical.[19] The measure has superior responsiveness in frailer populations to the Activities-specific Balance and Confidence Scale and is unlikely to suffer the ceiling effects apparent in the Falls Efficacy Scale.[21] Alternative methods of measurement, such as the Survey of Activities and Fear of Falling in the Elderly,[22] offer the potential to link fear of falling and activity restriction more closely but are too long and burdensome to consider as outcomes for clinical trials. The mFES[18] should be adopted as the standard against which new measures are developed and tested. It was recognized that the four items of instrumental activities of daily living in the mFES may require some modification. The ProFaNE group is undertaking further development work.

Recommendation 5: HRQoL

1. For the ProFaNE common outcome data set, the recommended measures of HRQoL are the Short Form 12 (SF12) version 2 and European Quality of Life Instrument (EuroQoL EQ-5D).

Rationale: Five measures were identified as having been used in fall-prevention trials previously. These were the Quality of Life Questionnaire of the European Foundation for Osteoporosis, the Short Form 36 (SF36), the Nottingham Health Profile, the Sickness Impact Profile, and the SF12 (reviewed in [23]). Although not used in fall-prevention trials previously, the group considered that the EuroQoL EQ-5D[24] should also be taken into account. The Quality of Life Questionnaire of the European Foundation for Osteoporosis was rejected because it is a disease-specific instrument that targets osteoporosis. The SF36, the Nottingham Health Profile, and the Sickness Impact Profile were rejected because, despite good psychometric properties, they place a substantial burden on respondents. The EuroQoL EQ-5D is simple and responsive to large changes in health and has been translated into several European

languages. It has been used widely in older populations.[23] Health utility indices have been developed allowing easy linkage to economic analysis.[25] There is some evidence of ceiling effects at item level, and the instrument may be insensitive to subtle changes in healthy populations,[23] but in frail populations, the EuroQoL EQ-5D demonstrates better sensitivity than the SF36.[26] The SF12 version 2 is a shortened version of the SF36. It has been translated widely and is practical. A health utility index is available for use alongside the SF12 in health economic analysis,[27] but it has a shorter track record than the EuroQoL EQ-5D utility. The SF12 has the potential to yield a broader assessment of mental and physical health than the EuroQoL EQ-5D in a response format that may be more appropriate in older people.[28] A head-to-head comparison is needed to substantiate these claims. An advantage of the EuroQoL EQ-5D is that, unlike the SF12, there is no license fee.

Recommendation 6: Physical Activity Measures

1. Further research is required before a measure of physical activity can be recommended for inclusion in the common data set.

Rationale: Systematic reviewing revealed seven measures that had previously been used in clinical trials.[3] The group considered that all these measures were too long and complex for routine administration. The experts did not identify any other measures as being potentially relevant. The Seven-Day Recall Questionnaire was considered the best basis for future development but will require modification for frailer populations.

Recommendation 7: Time Points for Follow-Up for the ProFaNE Common Data Set

1. Many fall-prevention interventions require longer-term follow up (12 months) because they have a delayed effect, taking time and compliance to evidence an effect.

Recommendation 8: Methodological Questions that Need to Be Addressed

1. Comparison of the performance of various HRQoL measures in clinical and cost-effectiveness analyses.
2. Development and validation of a simple physical activity measure.
3. Identification of cognitive ability required to ensure adequate completion of the various instruments.
4. Assessment of the practicality, reliability, and validity of proxy and postal response.
5. Consensus on the minimally clinically important differences for each of the common outcome data set measures.

DISCUSSION

ProFaNE is an extensive, international network of researchers who have collaborated to develop a common outcome data set for use in future trials and to identify important methodological questions. Further research is required to determine whether fall-prevention interventions can reduce serious injuries and to identify factors important to the successful implementation of programs. High-quality evidence could be gained from large-scale trials. A trial with serious injury as an end-point would likely require multiple sites (possibly multinational). Prospective pooling of individual patient data or trial results in meta-analysis would assist in yielding important evidence. All approaches would benefit from common outcomes.

International consensus can be gained using a variety of means.[4] The modified nominal group technique allows face-to-face discussion and resolution of concerns. Useful techniques to protect against inappropriate influence by individuals with strong opinions include group facilitation by an outside individual and a requirement to use evidence briefings in the consensus process. Frailty and Injuries: Cooperative Studies of Intervention Techniques undertook the first attempt to standardize definitions and measurement of outcomes for fall prevention trials.[29] The ProFaNE recommendation builds on this work by engaging international collaboration in identifying measures and integrating developments over the last 10 years.

Nearly all community-dwelling older people should be able to complete the outcome measurements. Apart from a measure of physical activity, the data set is sufficient to capture the important outcomes of fall injury trials. The common outcome data set is much shorter than the Frailty and Injuries: Cooperative Studies of Intervention Techniques data set and is of a length and format that may facilitate good response rates in postal follow-up.[7] It is anticipated that researchers will supplement the data set with other measures, reflecting local priorities and needs. It is also recognized that considerably more work needs to be undertaken to translate the measures into a range of languages and to ensure cultural compatibility.

These recommendations are intended for use in trials in which the target of intervention is a reduction in fall-related injury. Trials that target bone health or investigate the combined effects of fall and bone health interventions as a method of preventing nonvertebral fractures should use the recommendation. There is little congruity for the measures traditionally used in the two fields, and the broader effects of bone health interventions on falls, self-efficacy, and physical activity are, in the most part, not known.[30]

The authors are confident that the search strategy captured all measures that should have been considered. No information relating to thresholds for minimal clinically important differences for any of the psychological, HRQoL, or physical activity measures, from the individual or societal perspective, was found. This matter will be addressed in future revisions of the recommendations.

International collaboration has the potential to yield important information on the effectiveness of interventions and the cultural and psychosocial factors that influence their implementation. It is the authors' intention to refine the recommendations in 2007 as new evidence for the psychometric and practical properties of measures emerges.

Outcome Consensus Group

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